National Institute of Technology, Kurukshetra
Department of Robotics & Industrial Engineering
Theory Examination (End-Term)

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Question Paper

Month and Year of the Examination: **Nov / Dec 2025 (08-12-2025)**
Programme: B. Tech. (Robotics & Automation) Semester: Third (3rd) Sem.
Subject: **Introduction to Automation** Max. Marks: 50
Course Code: **RAPC-203** Time allowed: 03 hours

Note: Attempt ALL FIVE question. State clearly the assumptions made wherever required.

Que.1: Design the Automation for 'automatic coffee vending machine' using USA principles with special emphasis on components of automation. Use flow chart for explanation. (10 marks)

Que.2-a): What are the basic categories of the Sensors? Explain the performance terminologies of the sensors in context of their use. (06 marks)

Que.2-b): Write short note on Electric actuators. (04 marks)

Que.3-a): What are Industrial Control Strategies? Explain them in brief. P, PID ---? (05 marks)

Que.3-b): Elaborate the Discrete and Analog PLC system. (05 marks)

ue.4-a): Classify the Additive Manufacturing according to ASTM and elaborate their industrial application. (08 marks)

ue.4-b): What are the key technologies enables in Industry 4.0? (02 marks)

e.5-a): How will you justify and relate the role of SCADA and RTU in Real-time Monitoring? (04 marks)

e.5-b): Justify the adoption of digital twins and virtual commissioning in Industry 4.0. (03 marks)

e.5-c): Why machine vision system are considered as eyes of robots? State their applications. (03 marks)

---------- End of Question paper ----------

**********All The Best**********